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Calibration, not architecture, limits cross-subject wearable stress detection: a multi-dataset, decision-focused evaluation with label-efficient recalibration

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Abstract

Background Continuous wearable stress monitoring could support early intervention and self-management, but a stress estimate is useful only if it can guide a decision, which requires calibrated probabilities with demonstrable decision value, not discrimination. Most wearable stress studies have emphasised discrimination and have rarely reported calibration or decision-curve net benefit, the analyses that show whether a model is clinically useful for a chosen action threshold.

Methods Using three public Empatica E4 datasets (a laboratory benchmark, WESAD; a real-world occupational dataset, Nurse; and an induced acute-stress dataset, Exercise-Stress), we evaluated cross-subject stress detection using a leakage-controlled leave-one-subject-out protocol. A handcrafted-feature gradient-boosting baseline was assessed on all three datasets, and a compact 1D convolutional network, with and without self-supervised pre-training on the Nurse dataset. We quantified discrimination, probabilistic calibration (expected calibration error, Brier score, calibration slope and intercept, Spiegelhalter's Z), global and few-shot per-subject recalibration, and decision-curve net benefit, with label-permutation and random-split controls, and paired McNemar and DeLong tests.

Results Discrimination was saturated on WESAD (per-subject AUROC 0.99) but only modest on Nurse (0.60; 95% CI 0.52–0.68) and Exercise-Stress (0.59; 0.51–0.67). Neither deep variant surpassed the baseline (pooled AUROC 0.63 vs. 0.56 vs. 0.51). The calibration slope was below one on every dataset (0.54 on WESAD, 0.45 on Nurse, 0.22 on Exercise-Stress), the per-subject calibration error reached 0.22–0.29 on the real-world data, and a single global recalibration improved calibration in only 38% of the subjects. Few-shot per-subject recalibration reduced the Nurse calibration error from 0.22 to 0.07 when calibration windows were sampled across the recording, but this did not survive a prospective chronological protocol. In decision-curve terms, recalibration raised the mean incremental net benefit from –0.006 to +0.094 (robust to subject-level resampling) and the fraction of decision-useful thresholds from 58% to 79% (not robust to resampling).

Conclusions Calibration, not architecture, is the binding constraint on decision-useful cross-subject wearable stress detection for the compact models evaluated in this study; few-shot per-subject recalibration improves calibration when labels are sampled across the recording, though this benefit does not transfer to a prospective chronological setting. For decision support this means that a model with acceptable discrimination can still offer

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no clinical net benefit until it is calibrated to the individual, and that collecting a small subject-specific calibration set is a low-burden route to trustworthy stress probabilities. To our knowledge, this is the first application of decision-curve analysis to wearable-stress-detection.

Keywords Wearable sensors · Psychological stress · Photoplethysmography · Model calibration · Decision curve analysis · Cross-subject generalization · Label efficiency · Reproducibility

Abbreviations

AUROC	Area under the receiver operating characteristic curve
ECE	Expected calibration error
LOSO	Leave-one-subject-out
DCA	Decision-curve analysis
NB	Net benefit
PPG	Photoplethysmography
EDA	Electrodermal activity
BVP	Blood-volume pulse
HRV	Heart-rate variability
SDNN	Standard deviation of NN intervals
RMSSD	Root mean square of successive differences
TSST	Trier Social Stress Test
SSL	Self-supervised learning
CNN	Convolutional neural network
GB	Gradient boosting
CI	Confidence interval

Background

Psychological stress is a recognised contributor to cardiovascular, metabolic and mental-health burden, and its timely detection could support early intervention and self-management [1]. Wrist-worn devices that record photoplethysmography (PPG), electrodermal activity (EDA), skin temperature and accelerometry now make continuous, ambulatory stress monitoring feasible [2–4]. A stress estimate, however, is only useful insofar as it informs a decision; this implies that a deployable model must both *discriminate* stress from non-stress and output *calibrated* probabilities that can be trusted at a chosen action threshold [5].

Two properties make this difficult to achieve. Physiological responses to stress vary between individuals and devices, so a model may generalise poorly to unseen subjects [6]; and labelled data are scarce, motivating self-supervised and foundation models for physiological signals [7–11]. However, evaluation often lags behind: reported accuracies are high, leave-one-subject-out (LOSO) evaluation is applied inconsistently, and discrimination dominates the reporting [3].

Discrimination is insufficient for decision support. A model can rank stressed and non-stressed windows well yet be poorly *calibrated*, and using such probabilities at a fixed threshold can be of no benefit, or harmful, relative to trivial strategies [12]; modern networks are known to require recalibration [13]. In wearable stress detection, "calibration" frequently denotes per-user personalisation rather than probability calibration, a distinction we make explicit. To assess novelty we searched PubMed, Scopus, IEEE Xplore and Web of Science on 11 June 2026 (queries and counts in Additional file 1). A broad search confirmed a large, active literature (382, 14,058 and 24,589 records in PubMed, Scopus, and IEEE Xplore). Restricting to records also mentioning decision-curve analysis or net benefit yielded 91 candidates; on screening, none applied decision-curve analysis to a wearable

stress-detection model—the candidates addressed other outcomes such as cardiovascular, sleep and oncological prediction [14]. Within the stress-detection literature, evaluation remains centred on discrimination, and calibration and decision-curve analysis are rarely reported. To the best of our knowledge, this is the first application of decision curve analysis to wearable stress detection.

The contributions of this study are as follows:

- A leakage-controlled, multi-dataset LOSO evaluation of cross-subject wearable stress detection with explicit negative controls.
- Evidence that a calibrated feature baseline matches or exceeds a compact 1D-CNN with self-supervised pretraining across labelling budgets—an honestly reported negative result.
- A systematic assessment showing that probabilistic calibration degrades under subject shift and that a single global recalibration does not transfer across subjects.
- A label-efficient, few-shot per-subject recalibration that restores calibration under random sampling, together with a chronological analysis showing that this benefit does not transfer to a prospective protocol because of within-recording temporal drift.
- A decision-curve analysis linking calibration to clinical net benefit.

Related work

Wearable stress detection: datasets and models

Public multimodal datasets have driven the field, most prominently WESAD [15], with more recent real-world and protocol-controlled corpora including occupational recordings of hospital nurses [16] and induced acute-stress sessions [17]. Methods span handcrafted physiological features with classical classifiers [18, 19] and end-to-end deep networks [20], with reported accuracies on controlled benchmarks often high.

Self-supervised and foundation models

Self-supervised pretraining and PPG foundation models aim to learn transferable representations from unlabelled corpora [7, 8], [9], [10]; recent work applies universal time-series models for stress monitoring [11]. Evaluation, however, has focused on discrimination, and behavior under strict subject-independent protocols with limited per-subject data is less characterised.

Calibration and decision-analytic evaluation

Probabilistic calibration is central to clinical prediction [12], summarised by reliability diagrams, expected calibration error and the Brier score and assessed with the calibration slope/intercept [21] and tests of calibration-in-the-large [22]; networks benefit from post-hoc recalibration [13, 23, 24]. Decision curve analysis quantifies net benefit across decision thresholds [5, 25–27] and is now common in clinical prediction, yet it is rarely reported in wearable stress.

Reporting and reproducibility

Transparent-reporting guidance for prediction models includes machine-learning extensions [28, 29]; leakage between training and test partitions is a recurrent source of optimism that subject-independent protocols and paired tests (McNemar; DeLong [30]) are designed to detect.

Table 1 Notation

Symbol	Meaning
$\mathbf{x}_i$, y_i, s_i	window, binary stress label, subject index
S, N	number of subjects, number of windows
$\phi(\mathbf{x}) \in \mathbb{R}^d$	feature vector ($d = 11$ or 18)
f_θ, g_ψ, c_φ	encoder, SSL decoder, classifier head
$\hat{p}$, $\tilde{p}$	predicted, recalibrated probability
r, ρ	label fraction, SSL masking ratio
k, R	few-shot calibration size, repeats
M	calibration bins ($M = 10$)
α, β	calibration intercept, slope
π, p_t	prevalence, decision threshold

Table 2 Hyperparameters and protocol settings, from the released code

Component	Setting	Value
Windowing	length / overlap	60 s / none
Sampling	WESAD wrist / Nurse	64 Hz / 32 Hz
Normalisation	per-subject robust	median/IQR
Gradient boosting	stages M	100
	max depth	3
	learning rate ν	0.1
1D-CNN	blocks / widths	4 / 64, 128, 128, 128
	kernels / stride	7, 5, 5, 3 / 2
	embedding dim	128
SSL	mask ratio / optimiser / epochs / batch	0.5 / AdamW 10^{-3} / 50 / 256
Fine-tuning	optimizer / epochs / batch	AdamW $3{\times}10^{-4}$ / 30 / 128
Label fractions r	efficiency curve	$\{0.05, 0.10, 0.25, 1.00\}$
Calibration bins M	ECE	10
Few-shot recal.	sizes k / repeats R	$\{0, 5, 10, 20, 40\}$ / 30
Statistics	significance α	0.05 (two-sided)
Reproducibility	seed	1337

Methods

Overview and notation

We frame stress detection as binary classification of fixed-length windows of wrist-worn signal. A dataset is $\mathcal{D} = \{(\mathbf{x}_i, y_i, s_i)\}_{i=1}^N$, with windowed multichannel signal $\mathbf{x}_i$, binary label $y_i \in \{0, 1\}$ and subject index s_i. A model outputs $\hat{p}_i \in [0, 1]$. Three pathways are reported with different scope: a handcrafted-feature gradient-boosting baseline on all three datasets; a 1D-CNN (with/without self-supervised pretraining) on the Nurse dataset only; and calibration and decision-analytic evaluation with global and few-shot recalibration. Symbols are summarised in Table 1; settings in Table 2. Figure 1 shows the pipeline.

Datasets

3.2We used three public datasets recorded with the wrist-worn Empatica E4 [31] (Table 3). **WESAD** ($S = 15$) is a laboratory Trier Social Stress Test [32] benchmark [15]; the positive class is the stress condition and the negative class the union of baseline and amusement. **Nurse** is a real-world occupational dataset with self-report labels on a three-level scale [16]; we take level 2 as positive and level 0 as negative, discarding the intermediate level (a choice we verify in section "Deep and self-supervised models versus the baseline", where a three-class model cannot separate it), and exclude subjects with no negative windows, leaving 13 evaluable subjects. **Exercise-Stress**

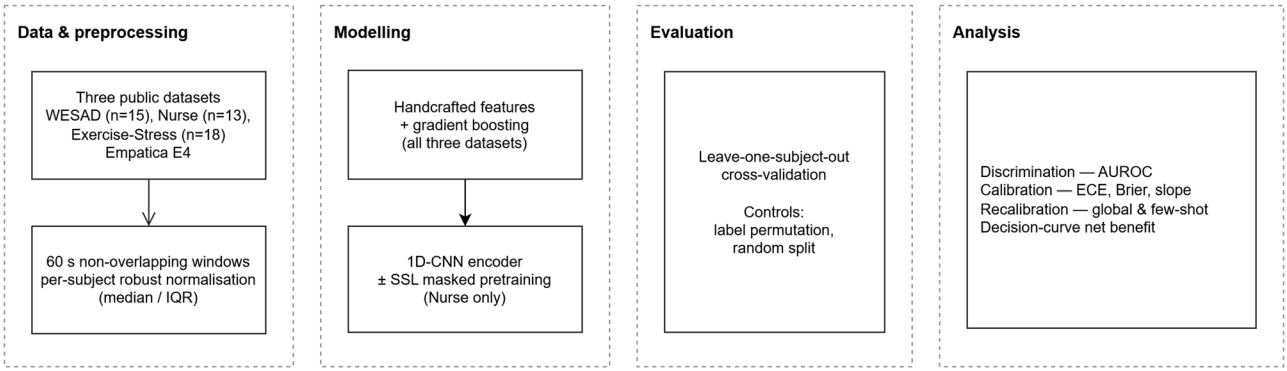


Fig. 1 Overview of the study pipeline. Three Empatica E4 datasets were windowed (60 s, non-overlapping) and per-subject normalized. A feature gradient-boosting baseline was evaluated on all three datasets; a 1D-CNN with optional self-supervised pretraining on the Nurse dataset. All models are assessed under leave-one-subject-out cross-validation with label-permutation and random-split controls, and characterised by discrimination, calibration, global and few-shot recalibration, and decision-curve net benefit

Table 3 Datasets. Folds are subjects with both classes; prevalence is the pooled fraction of stress windows

Dataset	Setting	Device	Folds	Positive vs negative	Prev.
WESAD	laboratory (TSST)	E4 wrist	15	stress vs baseline∪amusement	0.30
Nurse	real-world (hospital)	E4 wrist	13	self-report 2 vs 0	0.77
Exercise-Stress	induced (cognitive)	E4 wrist	18	stressor vs rest blocks	0.65

Table 4 Per-subject descriptive statistics (analysed windows)

Dataset	Windows/subject: median [IQR] (min–max)	Stress prevalence: median (range)
WESAD	37 [36–37] (35–38)	0.30 (0.29–0.32)
Nurse	308 [259–473] (132–766)	0.87 (0.15–0.97)
Exercise-Stress	20 [19–21] (18–24)	0.67 (0.58–0.68)

[17] uses the protocol variant with an unambiguous tag-to-block mapping ($S = 18$); the positive class comprises cognitive-stressor blocks and the negative class the remaining blocks. Feature dimensionalities are not harmonised across datasets, so all analyses are within-dataset and no cross-dataset transfer is performed. Across datasets the target is a momentary stress state, induced by protocol (WESAD, Exercise-Stress) or self-reported (Nurse); it is a transient physiological or perceived-stress label, not a clinical diagnosis of a stress-related disorder.

Per-subject window counts and class balance varied markedly (Table 4): WESAD and Exercise-Stress were uniform, whereas Nurse was highly heterogeneous (median 308 windows, range 132–766; subject-level stress prevalence 0.15–0.97 against a pooled 0.77). The small per-subject window count on Exercise-Stress (about 20) underlies the infeasibility of few-shot recalibration there, and the wide subject-level prevalence on Nurse underlies the failure of a single global recalibration to transfer across subjects.

Preprocessing and windowing

Signals are segmented into non-overlapping windows of $W = 60$ s (overlap is avoided as it can inflate leakage). Each channel is standardised with robust per-subject statistics computed over that subject's own recording,

$$\tilde{v}_{c,t}^{(s)} = \frac{v_{c,t}^{(s)} - \mathrm{med}_c^{(s)}}{\mathrm{IQR}_c^{(s)}}, \qquad \mathrm{IQR}_c^{(s)} = Q_{75}\big(v_{c,\cdot}^{(s)}\big) - Q_{25}\big(v_{c,\cdot}^{(s)}\big), \tag{1}$$

a zero interquartile range was replaced by one. Because Eq. (1) uses neither labels nor the train/test partition, it does not transfer target information across the held-out-subject boundary. It does, however, use the whole held-out recording; a causal variant that estimates these statistics from early windows only is examined in § "Calibration under subject shift", and affects calibration rather than discrimination. Robust normalization is applied to the deep pathway inputs (WESAD wrist; nurse) and to the signals from which WESAD features are computed; the Nurse and Exercise-Stress feature pathways operate on raw channel statistics, as the tree ensemble is invariant to monotone per-feature rescaling.

Feature representation

For a channel signal $\{v_1, \ldots, v_L\}$ we compute the mean, standard deviation and ordinary-least-squares slope,

$$m = \frac{1}{L}\sum_l v_l, \tag{2}$$

$$sd = \sqrt{\frac{1}{L}\sum_l (v_l - m)^2}, \tag{3}$$

$$\beta_1 = \frac{\sum_l (l - \bar{l})(v_l - m)}{\sum_l (l - \bar{l})^2}. \tag{4}$$

For acceleration we use the per-sample magnitude

$$a_l = \sqrt{a_{x,l}^2 + a_{y,l}^2 + a_{z,l}^2}. \tag{5}$$

The common feature set (Nurse, Exercise-Stress; $d = 11$) comprises the mean and standard deviation of heart rate, the mean/standard-deviation/slope of EDA and of skin temperature, and the mean/standard-deviation/slope of acceleration magnitude. For WESAD ($d = 18$) we additionally derive, from blood-volume-pulse peaks detected with a minimum spacing of 0.4 s and the resulting inter-beat intervals $\{\tau_p\}$,

$$\mathrm{HR} = \frac{60}{\bar{\tau}}, \quad \mathrm{SDNN} = \mathrm{std}(\tau_p), \quad \mathrm{RMSSD} = \sqrt{\overline{(\tau_{p+1} - \tau_p)^2}}, \tag{6}$$

set to zero when fewer than four peaks are detected, and the mean/standard deviation/slope of EDA, temperature and the three acceleration axes ($3 + 5 \times 3 = 18$). The two representations differ because WESAD's higher-rate wrist BVP (64 Hz) supports heart-rate-variability and per-axis acceleration features that the 1 Hz HR channel exported for the Nurse and Exercise-Stress recordings does not; we retain the richer set for WESAD but verify in § "Cross-subject discrimination" that the matched 11-feature set leaves the WESAD result unchanged.

Models

The primary model is a gradient-boosted decision-tree classifier [33] on ϕ, an additive ensemble

$$F_M(\phi) = \sum_{m=1}^{M} \nu\, \gamma_m\, h_m(\phi), \qquad \hat{p} = \sigma\big(F_M(\phi)\big), \tag{7}$$

with $\sigma(z) = (1 + e^{-z})^{-1}$, using $M = 100$ stages, depth-3 trees and learning rate $\nu = 0.1$. The deep encoder f_θ maps a normalised raw window $\mathbf{x} \in \mathbb{R}^{C \times T}$ to a 128-dimensional embedding through four convolution–batch-normalisation–GELU blocks (widths 64, 128, 128, 128; kernels 7, 5, 5, 3; stride 2),

$$(\mathbf{W} * \mathbf{u})_{o,t} = b_o + \sum_{c} \sum_{\tau=0}^{\kappa-1} W_{o,c,\tau}\, u_{c,t+\tau}, \tag{8}$$

followed by global average pooling. The encoder is optionally pretrained with a masked-reconstruction objective [34]: time steps are masked independently at ratio $\rho = 0.5$ and a four-layer transposed-convolution decoder g_ψ reconstructs the masked values,

$$\mathcal{L}_{\mathrm{SSL}}(\theta, \psi) = \frac{1}{|\mathcal{M}|} \sum_{t \in \mathcal{M}} \left\| g_\psi\big(f_\theta(\tilde{\mathbf{x}})\big)_{:,t} - \mathbf{x}_{:,t} \right\|_2^2, \tag{9}$$

optimized with AdamW [35] (learning rate 10^{-3}, weight decay 10^{-4}) for 50 epochs at a batch size 256. The pretraining pool comprises the unlabelled windows of all subjects, including the held-out subject (labels are never used); we also evaluated a stricter variant that excludes the held-out subject from pretraining (section "Deep and self-supervised models versus the baseline"). Because this gives the self-supervised model access to the held-out subject's unlabelled signal distribution, it biases the comparison in its favor; therefore, its failure to exceed the baseline is a conservative result. A head c_φ ($128 \to 64 \to 1$, GELU) produces the logit z, and f_θ, c_φ are fine-tuned by binary cross-entropy

$$\mathcal{L}_{\mathrm{CLS}} = -\frac{1}{N} \sum_{i} [y_i \log \hat{p}_i + (1 - y_i) \log(1 - \hat{p}_i)], \tag{10}$$

with AdamW (learning rate 3×10^{-4}) for 30 epochs at a batch size 128. The *scratch* variant trains from random initialisation, the *SSL* variant from pretraining. For label efficiency, the classifier was trained with a subject-stratified fraction $r \in \{0.05, 0.10, 0.25, 1.00\}$ of the training labels. The deep pathway was evaluated using the Nurse dataset. These two models were chosen deliberately. Gradient boosting on handcrafted physiological features is a strong, standard baseline for small tabular datasets and the common reference in wearable stress detection; the compact 1D-CNN represents the end-to-end deep alternative, kept small to suit the limited per-subject data. Contrasting a calibrated feature baseline with a representative deep model isolates whether architecture or calibration is the binding constraint, which is the study's question, rather than surveying many architectures.

Evaluation protocol and controls

All models were evaluated using leave-one-subject-out (LOSO) cross-validation (Algorithm 1), and we report pooled and per-subject metrics. Two controls verify the absence of leakage: *label permutation* (training labels permuted; a leakage-free pipeline yields AUROC near 0.5) and *random split* (windows partitioned ignoring subject; a far higher score indicates that LOSO difficulty stems from between-subject variation). Discrimination is the area under the ROC curve,

$$\mathrm{AUROC} = \frac{1}{N_+ N_-} \sum_{i:y_i=1} \sum_{j:y_j=0} \left[\mathbb{1}(\hat{p}_i > \hat{p}_j) + \tfrac{1}{2} \mathbb{1}(\hat{p}_i = \hat{p}_j) \right]. \tag{11}$$

Calibration

Predictions are partitioned into $M = 10$ equal-width bins [36]; with bin confidence and accuracy $\mathrm{conf}(B_m), \mathrm{acc}(B_m)$,

$$\mathrm{ECE} = \sum_{m} \frac{|B_m|}{N} |\mathrm{acc}(B_m) - \mathrm{conf}(B_m)|, \tag{12}$$

$$\mathrm{BS} = \frac{1}{N}\sum_i (\hat{p}_i - y_i)^2. \tag{13}$$

The Brier score and calibration slope/intercept were binning-independent. We fit the logistic calibration model

$$\mathrm{logit}(\Pr(y=1)) = \alpha + \beta\ \mathrm{logit}(\hat{p}), \tag{14}$$

($\beta < 1$ indicates over-confidence) and test calibration-in-the-large with Spiegelhalter's statistic

$$Z = \frac{\sum_i (y_i - \hat{p}_i)(1-2\hat{p}_i)}{\sqrt{\sum_i (1-2\hat{p}_i)^2 \hat{p}_i(1-\hat{p}_i)}}. \tag{15}$$

Calibration metrics were reported both pooled and as the mean over per-subject values, the latter being primary for the subject-shift analysis.

Recalibration

We recalibrate by logistic (Platt) scaling,

$$\tilde{p} = \sigma(a\ \mathrm{logit}(\hat{p}) + b), \tag{16}$$

fitted near-unregularised. In the *global* regime (a, b) are fitted on the training-fold predictions; in the *few-shot* regime they are fitted on $k \in \{0, 5, 10, 20, 40\}$ labelled windows of the held-out subject and the metric averaged over $R = 30$ random draws (Algorithm 3), skipping folds with fewer than $\max(k, 5)$ windows or a single class. We use logistic scaling because it corrects both the calibration slope and intercept with two parameters, which suits the small per-subject calibration sets; we compare it against temperature scaling, isotonic regression and beta calibration in § "Label-efficient few-shot recalibration".

Decision-curve analysis

In this setting, the decision is whether to act on a window, for example, to raise a stress alert or trigger a brief check-in or intervention. A true positive is a correctly flagged stress window, and a false positive is an unnecessary alert. The threshold p_t encodes their relative cost through the odds $p_t/(1-p_t)$, so that $p_t = 0.2$ treats one missed stress window as costly as four unnecessary alerts. Net benefit places true and false positives on this common scale: at threshold p_t a window is acted upon if $\tilde{p} \geq p_t$, and

$$\mathrm{NB}(p_t) = \frac{\mathrm{TP}(p_t)}{N} - \frac{\mathrm{FP}(p_t)}{N}\frac{p_t}{1-p_t}, \tag{17}$$

compared with treat-all and treat-none references

$$\mathrm{NB}_{\mathrm{all}}(p_t) = \pi - (1-\pi)\frac{p_t}{1-p_t}, \qquad \mathrm{NB}_{\mathrm{none}}(p_t) = 0. \tag{18}$$

A model is decision-useful when $\mathrm{NB}(p_t) > \max(\mathrm{NB}_{\mathrm{all}}(p_t), 0)$; we report the incremental net benefit and the fraction of thresholds on a grid from 0.05 to 0.95 (steps of 0.05; 19 thresholds) at which this holds for the uncalibrated and recalibrated scores (Algorithm 4).

Statistical analysis

Paired classifications are compared with McNemar's test on the discordant counts b, c, using the exact binomial test for fewer than 25 discordant pairs and otherwise the continuity-corrected statistic

$$\chi^2 = \frac{(|b-c|-1)^2}{b+c}, \tag{19}$$

and correlated AUROCs with the fast DeLong method,

$$Z = \frac{\mathrm{AUROC}_A - \mathrm{AUROC}_B}{\sqrt{\mathrm{Var}(\mathrm{AUROC}_A - \mathrm{AUROC}_B)}}. \tag{20}$$

Pooled AUROC intervals use the DeLong variance; per-subject metrics use the normal-approximation interval over the n_S folds, $\bar{m} \pm 1.96\,\mathrm{sd}(m)/\sqrt{n_S}$, with all tests two-sided at $\alpha = 0.05$. Paired per-subject comparisons used the Wilcoxon signed-rank test on per-fold AUROCs, with the sign test as a robustness check. Uncertainty for calibration and decision-curve quantities was estimated by resampling subjects with replacement (5000 replicates), since windows within a subject are correlated.

Implementation and reproducibility

Computational cost was modest for both models on a single T4 GPU (largest Nurse LOSO fold, 4805 training and 766 test windows). The feature and gradient-boosting baseline was smaller and faster than the 1D-CNN (0.14 vs 0.75 MB; 3.4 vs 25.8 s to train; 0.002 vs 0.05 ms per window at inference), so the calibrated baseline is also the lighter deployment option; the deep model's limitation is accuracy rather than cost. Feature extraction, gradient boosting, and logistic recalibration used scikit-learn [37]; the 1D-CNN and pretraining used PyTorch [38]. The DeLong, Spiegelhalter, and calibration routines were unit-tested against synthetic data. Seeds are fixed (seed 1337); experiments were performed on the Kaggle platform with an NVIDIA T4 GPU. We report in line with the TRIPOD+AI recommendations [29]; code and the scripts that regenerate every figure and table are openly available at https://github.com/kmdr82/wearable-stress-calibration.The structured literature search is summarized in Table 5.

Table 5 Structured literature search (11 June 2026). The broad search establishes sensitivity; the decision-curve-restricted column is the screening pool; eligibility requires applying decision-curve analysis to a wearable stress-detection model

Database	Broad (stress × wearable)	+ DCA/net benefit	Eligible
PubMed	382	1	0
Scopus	14,058	89	0
IEEE Xplore	24,589	0	0
Web of Science	–	1	0
ACM DL (suppl.)	105	screened	0
Google Scholar (suppl.)	–	top-50 screened	0

Algorithm 1 Leave-one-subject-out training and evaluation

Require: dataset $\mathcal{D}$; model type τ; label fraction r; control mode
window and per-subject normalise (Eq. (1)); for the feature pathway compute ϕ (Eqs. (2)–(6))
for each subject s **do**
  $\mathcal{D}_{\text{tr}} \leftarrow \{i : s_i \neq s\}$; $\mathcal{D}_{\text{te}} \leftarrow \{i : s_i = s\}$
  if control = permutation **then** permute $\{y_i : i \in \mathcal{D}_{\text{tr}}\}$
  else if control = random **then** redraw splits ignoring s
  end if
  if τ = GB **then** fit F_M (Eq. (7))
  else
    if τ = CNN-SSL **then** $\theta \leftarrow$ SSLPretrain(all unlabelled windows) (Alg. 2)
    end if
    sample subject-stratified r-fraction; fit f_θ, c_φ by BCE (Eq. (10))
  end if
  $\hat{p}_i \leftarrow$ model prediction for $i \in \mathcal{D}_{\text{te}}$
end for
compute pooled and per-subject AUROC, ECE, Brier, slope/intercept, Z
return $\{\hat{p}_i\}$, metrics

Results

Pipeline integrity

Label permutation collapsed pooled discrimination to chance (WESAD AUROC 0.43; Nurse 0.48), and on WESAD a subject-agnostic random split reached AUROC 1.00 against a LOSO AUROC of 0.98, confirming that the controlled task is genuinely easy rather than inflated by split leakage.

Cross-subject discrimination

Discrimination differed markedly across datasets (Fig. 2; Table 6). On WESAD it was saturated (per-subject mean AUROC 0.99, 95% CI 0.98–1.00). On the less constrained datasets it was modest: the baseline reached 0.60

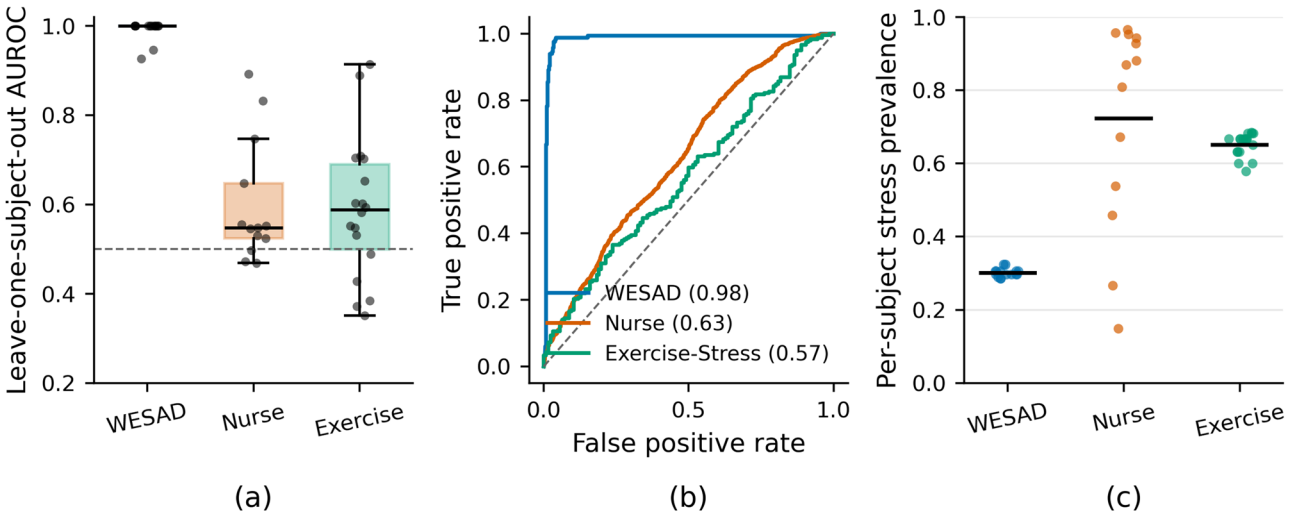


Fig. 2 Cross-subject discrimination. (**a**) Leave-one-subject-out AUROC per held-out subject with box plots. (**b**) Pooled ROC curves; the AUROC values shown are pooled and differ from the primary per-subject means in Table 6. (**c**) Per-subject stress prevalence (bars: dataset means)

Table 6 Cross-subject discrimination and leakage controls. The per-subject AUROC and its 95% CI (normal-approximation interval over per-fold AUROCs) are primary; the pooled AUROC is on concatenated held-out predictions

Dataset	Folds	Per-subject AUROC (95% CI)	Pooled	Per-fold range	Shuffled / Random
WESAD	15	0.992 (0.979–1.000)	0.983	0.927–1.000	0.43 / 1.00
Nurse	13	0.601 (0.517–0.684)	0.631	0.468–0.893	0.48 / –
Exercise-Stress	18	0.589 (0.510–0.668)	0.568	0.352–0.914	– / –

(95% CI 0.52–0.68) on Nurse and 0.59 (0.51–0.67) on Exercise-Stress. Between-subject variability was large (SD 0.14 and 0.16), the lower confidence limits approached chance, and individual subjects fell below chance (minimum per-fold AUROC 0.47 and 0.35). We report the pooled AUROC for completeness but treat its DeLong interval as anti-conservative under within-subject clustering; the subject-level interval is the honest summary. To check that the cross-dataset differences were not an artefact of the feature representation, we re-ran WESAD with the common 11-feature set used for Nurse and Exercise-Stress, changing only the feature set (Additional file; script `matched_features_wesad.py`). Discrimination and calibration were essentially unchanged (per-subject AUROC 0.992 vs 0.992 with 18 features; per-subject ECE 0.037 vs 0.036; Brier 0.027 vs 0.029; calibration slope 0.62 vs 0.54), confirming that the saturated WESAD result and the gap to the real-world datasets reflect the recording setting rather than the richer WESAD representation.

Algorithm 2 Self-supervised masked-reconstruction pretraining

Require: unlabelled training windows $\mathcal{X}$; mask ratio ρ
initialise f_θ, g_ψ
repeat
  sample minibatch; mask time indices at ratio ρ; form $\tilde{\mathbf{x}}$
  update (θ, ψ) by minimising $\mathcal{L}_{\text{SSL}}$ (Eq. (9))
until stopping
return θ

Algorithm 3 Few-shot per-subject recalibration

Require: held-out predictions $\{(\hat{p}_i, y_i) : i \in \mathcal{D}_{\text{te}}\}$; size k; repeats R
if $|\mathcal{D}_{\text{te}}| \le \max(k, 5)$ **or** single-class **then return** UNDEFINED
end if
for rep $= 1, \ldots, R$ **do**
  sample $\mathcal{K} \subset \mathcal{D}_{\text{te}}$, $|\mathcal{K}| = k$; **if** single-class **continue**
  fit (a, b) on $\{(\text{logit}\,\hat{p}_i, y_i) : i \in \mathcal{K}\}$ (Eq. (16))
  $e_{\text{rep}} \leftarrow$ ECE on $\mathcal{D}_{\text{te}} \setminus \mathcal{K}$ (Eq. (12))
end for
return $\text{mean}_{\text{rep}}\, e_{\text{rep}}$

Deep and self-supervised models versus the baseline

On Nurse, neither deep variant outperformed the baseline at any labelling budget (Fig. 3). At full labelling the pooled AUROC was 0.63 (gradient boosting), 0.56 (self-supervised 1D-CNN) and 0.51 (from-scratch 1D-CNN). At the subject level, the baseline exceeded the self-supervised model in 8/13 folds, but this difference was not significant under paired tests that respect subject clustering (Wilcoxon signed-rank $p = 0.59$; sign test $p = 0.58$) : the two models were statistically indistinguishable per subject (mean per-subject AUROC 0.60 vs 0.57). The from-scratch model was clearly inferior, both per subject (11/13 folds; Wilcoxon $p = 0.017$) and pooled (DeLong $p = 1.6 \times 10^{-22}$), with a mean per-subject AUROC of 0.49. Self-supervised pretraining improved training from scratch but did not exceed the calibrated baseline. We report the pooled tests (DeLong $p = 1.7 \times 10^{-7}$; McNemar $p = 2.5 \times 10^{-13}$) for completeness, but they are anti-conservative under within-subject clustering, and the subject-level paired tests are the primary inference. Excluding the held-out subject from self-supervised pretraining (a fully subject-independent protocol) did not change the conclusion: strict SSL reached pooled AUROC 0.533 and mean per-subject

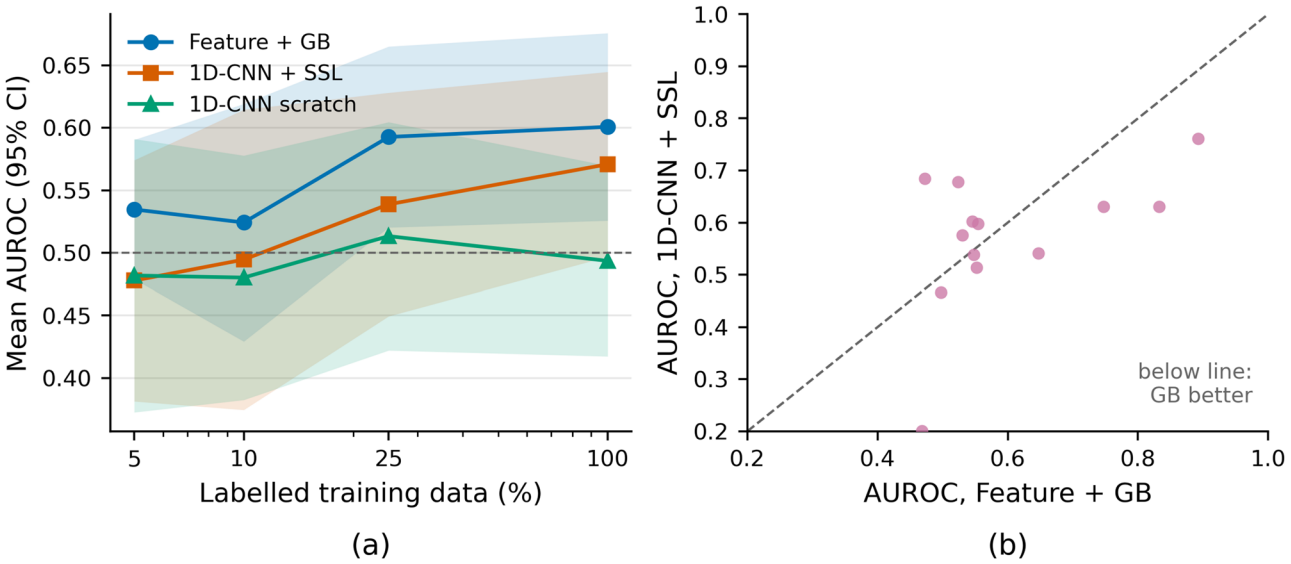


Fig. 3 Deep models versus the baseline (Nurse). (**a**) Mean AUROC (95% CI band) versus labelled-data fraction. (**b**) Per-subject AUROC of the self-supervised model versus the baseline; points below the identity favour the baseline

AUROC 0.531, below both the transductive variant (0.562) and the calibrated baseline (0.631/0.601; the baseline was higher in 10/13 folds, Wilcoxon $p = 0.08$). Transductive pretraining therefore gave the self-supervised model only a small, non-decisive advantage, and removing it leaves the negative result intact.

Algorithm 4 Decision-curve net benefit

Require: recalibrated $\{(\tilde{p}_i, y_i)\}$; threshold grid $\mathcal{T}$

$\pi \leftarrow \frac{1}{N}\sum_i y_i$
for $p_t \in \mathcal{T}$ **do**
  predict positive iff $\tilde{p}_i \geq p_t$; compute $\mathrm{NB}(p_t)$ (Eq. (17)), $\mathrm{NB}_{\mathrm{all}}(p_t)$ (Eq. (18))
  $\mathrm{inc}(p_t) \leftarrow \mathrm{NB}(p_t) - \max(\mathrm{NB}_{\mathrm{all}}(p_t), 0)$
end for
return {NB}, {inc}, fraction with inc > 0

Reinstating the discarded intermediate level and training a three-class model on Nurse did not yield a usable intermediate detector. The intermediate class was sparse (419 windows, 7%; present in only 5 of 15 subjects), and under LOSO it was essentially never predicted: one-vs-rest AUROC was 0.215 for the intermediate class (below chance), 0.624 and 0.584 for baseline and stress, macro-AUROC 0.474, and balanced accuracy 0.375, with the classifier collapsing to the majority stress class (78–91% of every true class predicted as stress). This supports the binary formulation used throughout.

Calibration under subject shift

The calibration slope was below one on *all* datasets (Table 7; Fig. 4). On Nurse the mean per-subject expected calibration error (ECE) was 0.22 (far larger than the pooled value of 0.085, which masks subject-specific miscalibration), with slope 0.45, intercept 0.55 and a strongly significant Spiegelhalter test ($Z = 16.0$). Exercise-Stress showed steeper overconfidence (mean ECE 0.29, slope 0.22, $Z = 8.6$). The saturated WESAD model had a small absolute calibration error (ECE 0.03) and near-zero intercept (−0.05), yet a slope of 0.54 and a significant test ($Z = 7.1$), indicating residual over-confidence on the logit scale that the ECE does not capture. Consistently, the Brier score of the uncalibrated real-world models was no better than (and on Exercise-Stress worse than) a no-information predictor of the prevalence ($\pi(1-\pi) = 0.178$ vs 0.176 on Nurse; 0.227 vs 0.258 on Exercise-Stress,

Table 7 Calibration (global, gradient boosting). ECE ($M = 10$) per-subject mean is primary, pooled in parentheses. Brier pooled; slope/intercept and Spiegelhalter Z on pooled predictions (all $p < 0.001$).

Dataset	ECE (per-subj/pooled)	Brier	Slope	Intercept	Z
WESAD	0.036 / 0.027	0.029	0.535	−0.050	7.08
Nurse	0.223 / 0.085	0.176	0.445	0.547	16.01
Exercise-Stress	0.289 / 0.175	0.258	0.219	0.474	8.59

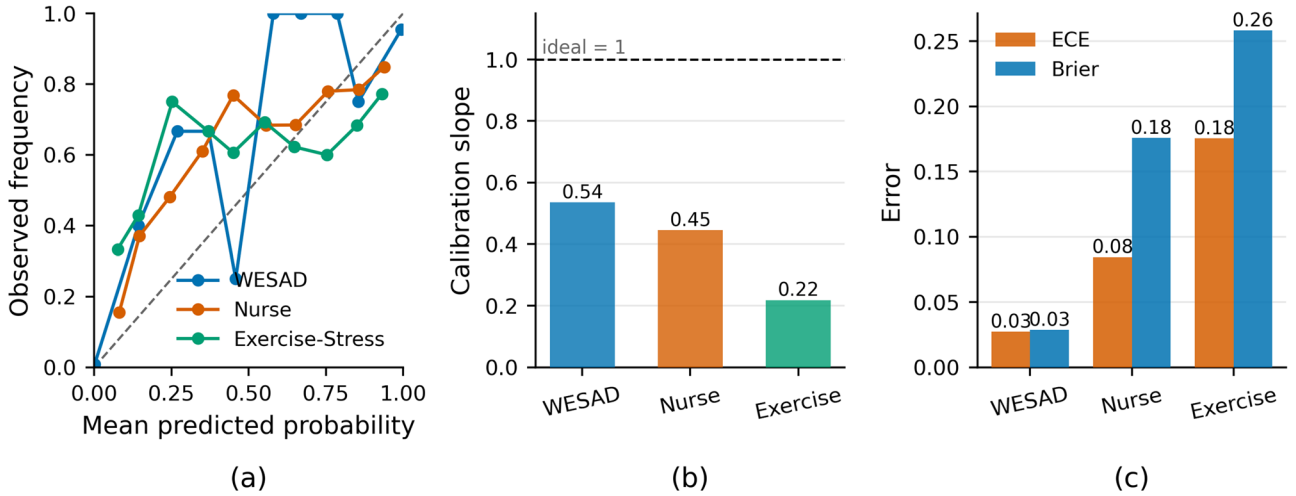


Fig. 4 Calibration under subject shift. (**a**) Reliability diagrams. (**b**) Calibration slope (dashed line: ideal value one). (**c**) ECE and Brier score

worse than the constant predictor), whereas on WESAD it was far lower (0.211 vs 0.029). On Nurse, a single global recalibration reduced the per-subject expected calibration error in only 5/13 (38%) of held-out folds, confirming that a single global map does not transfer across subjects. Because the pooled samples are large, the Spiegelhalter test is significant throughout; the effect sizes (ECE and slope) are substantive measures. Normalisation used per-subject median and IQR over the whole recording, which is not available prospectively. Recomputing these statistics from only each subject's earliest windows and applying them causally left discrimination essentially intact (per-subject AUROC 0.986, 0.947 and 0.928 for the first 50%, 20% and 10% of windows, versus 0.992) but degraded calibration markedly (per-subject ECE 0.106, 0.146 and 0.163, versus 0.036; Brier up to 0.134 from 0.029). Even on the saturated benchmark, calibration, not discrimination, is the quantity sensitive to this leakage. Subject-level bootstrap 95% intervals for the per-subject mean ECE were [0.018, 0.057] (WESAD), [0.135, 0.330] (Nurse) and [0.259, 0.319] (Exercise-Stress); the Nurse interval is wide, so its point estimate is less precise than the small pooled value suggests.

Label-efficient few-shot recalibration

When calibration windows were sampled at random from a held-out subject's recording, subject-specific recalibration reduced miscalibration (Table 8; Fig. 5). On Nurse the mean per-subject ECE fell from 0.223 ($k = 0$; 95% CI 0.136–0.329) to 0.150 ($k = 10$; 0.132–0.171), 0.094 ($k = 20$; 0.075–0.115) and 0.068 ($k = 40$; 0.051–0.086), with a minimal budget transiently harmful ($k = 5$: 0.257); the between-subject SD of ECE contracted from 0.18 to 0.04 , and the $k = 20$ and $k = 40$ intervals lay below the uncalibrated interval. On Exercise-Stress the procedure did not help: with about 20 windows per subject, small budgets worsened calibration and $k = 20$ left only 7 evaluable folds ($k = 40$ none), an explicit data-availability boundary. The benefit did not depend on the recalibration family. Repeating the few-shot procedure on Nurse with temperature scaling, isotonic regression and beta calibration

Table 8 Few-shot per-subject recalibration: mean ECE (between-subject SD) and evaluable folds versus k (k=0 is global recalibration)

k	Nurse: ECE (SD) [folds]	Exercise-Stress: ECE (SD) [folds]
0	0.223 (0.181) [13]	0.289 (0.069) [18]
5	0.257 (0.057) [13]	0.352 (0.054) [18]
10	0.150 (0.036) [13]	0.309 (0.032) [18]
20	0.094 (0.037) [13]	0.415 (0.078) [7]
40	0.068 (0.032) [13]	– [0]

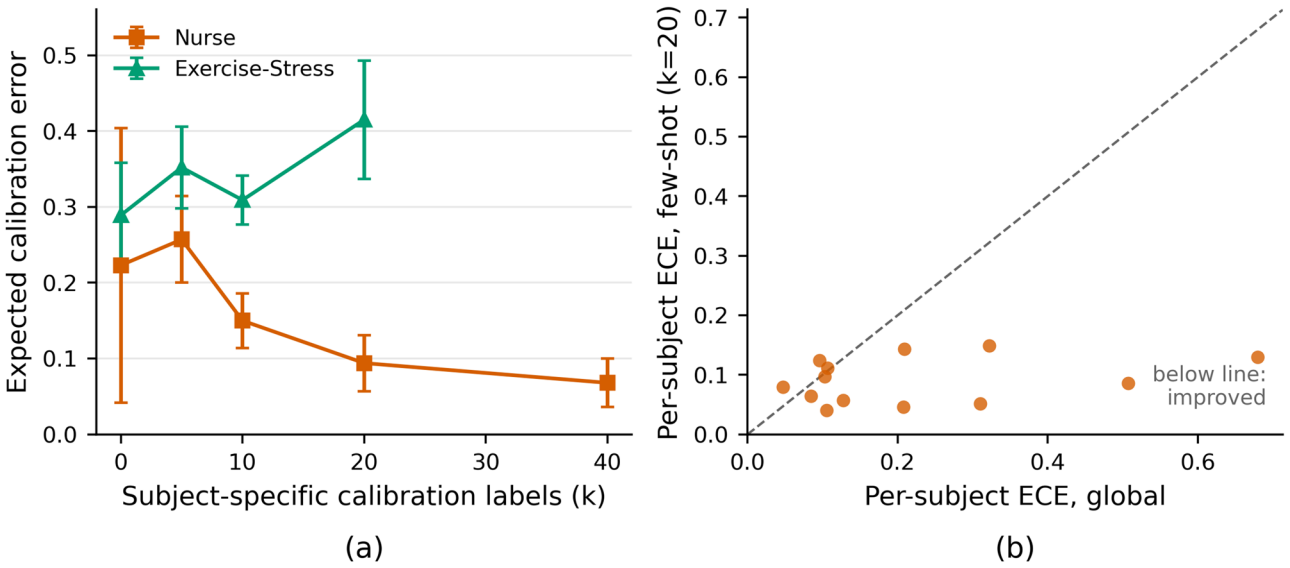


Fig. 5 Few-shot per-subject recalibration. (**a**) ECE versus the number of subject-specific labels k (mean±SD). (**b**) Per-subject ECE after few-shot recalibration (k = 20) versus the global baseline on Nurse; points below the identity indicate improvement

in place of logistic scaling reduced miscalibration in every case (mean per-subject ECE at k = 20: logistic 0.091 , isotonic 0.106, beta 0.127, temperature 0.152; uncalibrated 0.223). Logistic scaling was best; temperature scaling, which rescales only the logit slope, was worst, consistent with the large calibration intercept (0.55) that a slope-only map cannot correct.

Because random sampling places calibration windows throughout the recording, temporally interspersed with the evaluation windows, it can overstate what is achievable prospectively, when calibration labels must be collected before deployment. We therefore repeated the analysis chronologically, calibrating on each subject's earliest windows and evaluating on the later ones. Two obstacles appeared. First, calibration was frequently infeasible: a median of 47 windows (IQR 31–158; range 5–327) elapsed before both classes were observed, so the first $k \leq 40$ windows were single-class in most subjects and no calibration model could be fitted. Second, where an earliest class-balanced calibration block could be formed, recalibration did not improve calibration: the mean per-subject ECE was 0.324 (k = 5), 0.326 (k = 10), 0.293 (k = 20) and 0.225 (k = 40) over 11 evaluable folds, no better than the uncalibrated 0.223 and far above the randomly sampled values. Per-subject recalibration therefore lowers miscalibration only when calibration data are distributed across the deployment period; fitted on early data alone it does not transfer, indicating that within-recording temporal non-stationarity, not only between-subject shift, limits calibration.

Decision-curve analysis

Recalibration translated into a greater decision value (Fig. 6). Over a grid of thresholds from 0.05 to 0.95, the few-shot-recalibrated model exceeded the better of the treat-all and treat-none references at 79% of thresholds,

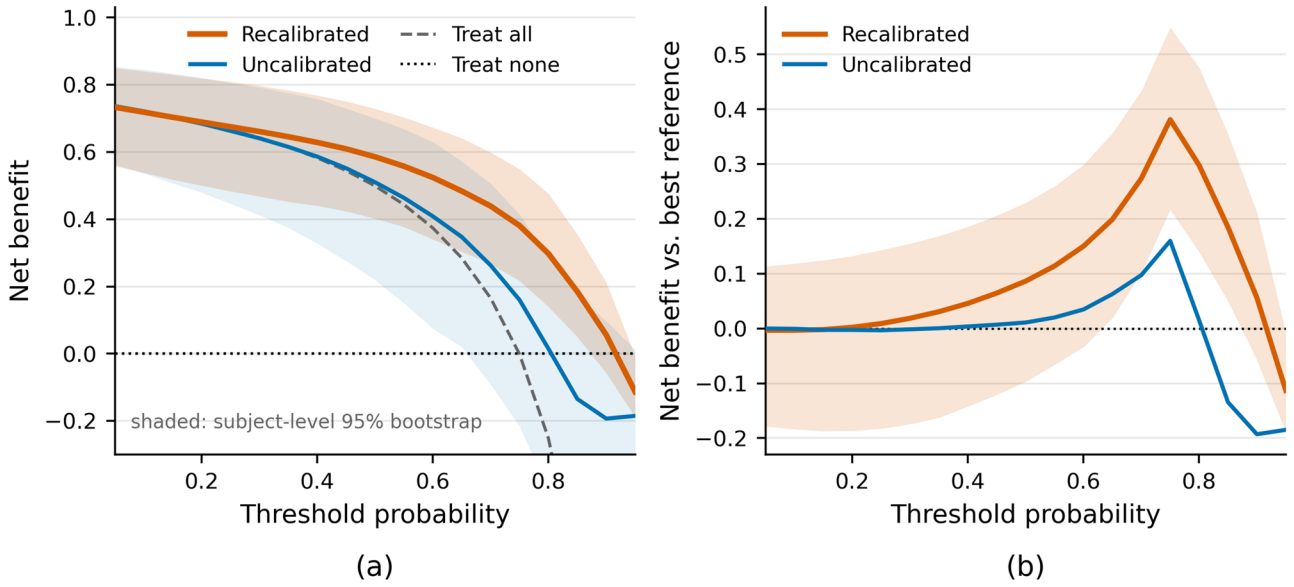


Fig. 6 Decision-curve analysis (Nurse). (**a**) Net benefit versus threshold for the uncalibrated and few-shot-recalibrated models with treat-all/treat-none references. (**b**) Net benefit relative to the best reference; positive values (shaded) indicate decision value. Shaded regions are subject-level 95% bootstrap bands (5000 replicates); their width shows that the net-benefit ranking is uncertain and that the decision-useful threshold fraction is not robust, while the incremental net benefit remains positive

versus 58% for the uncalibrated model. The mean incremental net benefit relative to the best reference was slightly negative for the uncalibrated model (−0.006) but clearly positive after recalibration, indicating that calibration, not discrimination, governed whether the model was decision-useful. Because this recalibration uses the same randomly sampled few-shot procedure, the decision-value gain is likewise an upper bound; under a chronological protocol the recalibration did not transfer (§4.5). The net benefit is on the scale of true positives per window, so an incremental value near 0.09 corresponds to roughly nine additional net true detections per 100 windows relative to the best default strategy, after the false-positive cost is subtracted at each threshold. Subject-level bootstrap (resampling subjects, 5000 replicates) gave a mean incremental net benefit of +0.094 (95% CI +0.004 to +0.161; difference from the uncalibrated model +0.100, CI +0.017 to +0.248), so the net-benefit gain is robust. The increase in the decision-useful fraction from 58% to 79% was not robust at the subject level (difference +0.21, CI −0.11 to +0.68), reflecting the small number of subjects; we therefore rely on the incremental net benefit rather than the threshold fraction.

Discussion

Principal findings

Across three Empatica E4 datasets evaluated under a leakage-controlled LOSO protocol, the practical bottleneck for decision-useful wearable stress detection was probabilistic calibration rather than model architecture. Discrimination was dataset-dependent (saturated on the controlled benchmark; modest elsewhere), a compact 1D-CNN with or without self-supervised pretraining did not outperform a calibrated baseline, calibration degraded under subject shift on all datasets, a single global recalibration did not transfer, and few-shot per-subject recalibration improved calibration and net benefit when calibration windows were sampled across the recording, but not prospectively from early data alone or where per-subject data were scarce.

Table 9 Representative wearable stress- and affect-detection studies versus this work. All report discrimination (accuracy, F1, or AUROC); to our knowledge none report probabilistic calibration or decision-curve net benefit. The calibration and decision-curve columns are consistent with the structured search of additional file 1, which found no application of decision-curve analysis to wearable stress detection

Study	Dataset(s)	Validation	Calibration	DCA
Schmidt et al. [15]	WESAD	LOSO	No	No
Sarkar & Etemad [7]	ECG/PPG affect corpora	LOSO	No	No
Hosseini et al. and Nurse classifiers [16]	Nurse	LOSO / random split	No	No
PPG foundation models [9, 10]	large PPG corpora	held-out subjects	No	No
This work	WESAD, Nurse, Exercise-Stress [17]	strict LOSO	**Yes**	**Yes**

Interpretation

Miscalibration was strongly subject-specific: pooled error understated per-subject error and a global map did not transfer, the expected consequence of per-subject physiological and base-rate shift, which a few subject-specific labels can correct. The cost was concrete: uncalibrated real-world models attained Brier scores no better than (and on exercise stress worse than) a no-information predictor and negative net benefit at higher thresholds. A sub-unit calibration slope, even on the saturated benchmark, shows calibration and discrimination as distinct axes. The absence of a deep advantage is consistent with the small per-subject samples and the strength of handcrafted features, and we interpret it as a property of this data and compute regime rather than a general claim. Improved calibration does not improve discrimination, since AUROC is invariant to any monotone recalibration, and it does not necessarily improve every threshold-specific decision: the subject-level uncertainty on the decision-useful fraction shows that not all thresholds benefit reliably.

Comparison with prior work

High accuracy on controlled benchmarks is common [2, 39], consistent with our saturated WESAD results. Gains from self-supervised and foundation models are typically shown at scale and for discrimination [11]; our negative result scopes rather than contradicts these. Calibration is well studied in clinical prediction [12] but seldom reported in wearable stress, and decision-curve analysis [5], though applied to wearable-derived prediction for other outcomes [14], has to our knowledge not been used for wearable stress detection. Table 9 places this work against representative studies: prior wearable stress detection reports discrimination under subject-dependent or subject-independent validation, but not probabilistic calibration or decision-curve net benefit, which this study adds.

Implications for practice and wearable data management

Subject-independent evaluation and the reporting of calibration and decision utility, not discrimination alone, should be standard, since a model with an acceptable AUROC can be net-harmful before recalibration. Datasets should retain subject identifiers and sufficient windows per subject: the few-shot remedy succeeded with hundreds of windows per subject but failed at approximately twenty. A small subject-specific labelled set (order of twenty windows) is a realistic, low-burden data-collection target for deployment, and the label provenance and prevalence should be documented. We frame usefulness as decision support: whether a calibrated stress probability can guide an alert or check-in at a chosen threshold, not as a diagnostic or regulatory clinical decision.

Strengths and limitations

The strengths include three datasets with explicit leakage controls, strict subject-independent evaluation with subject-level uncertainty, decision-focused analysis, transparent reporting [29], and open code. Limitations: the

datasets are small (15, 13, 18 subjects), intervals are wide, and some subjects fell below chance; labels differ in nature (self-report versus laboratory-induced) and the intermediate Nurse level was discarded, a choice supported by a three-class model that could not separate it (section "Deep and self-supervised models versus the baseline"); features were not harmonised across datasets (within-dataset analyses only); the deep model was compact owing to compute, and the self-supervised pretraining pool included the unlabelled windows of all subjects (labels never used), though a stricter subject-independent variant gave the same conclusion (section "Deep and self-supervised models versus the baseline"); the few-shot remedy lowers miscalibration only when calibration windows are sampled across the recording and does not transfer to a prospective chronological protocol; device diversity was limited to the wrist-worn Empatica E4; recalibration used post-hoc scaling families (logistic, temperature, isotonic, beta) rather than hierarchical or Bayesian subject-specific updating; and the pooled significance tests are anti-conservative under clustering, so we relied on subject-level intervals and per-subject majorities. Per-subject normalisation used whole-recording statistics; a causal variant lowered calibration quality while leaving discrimination high (section "Calibration under subject shift"). The labels denote momentary induced or self-reported stress rather than a clinical diagnosis, so the models detect a stress state and are not diagnostic instruments.

Future work

Future work should prospectively evaluate larger, multi-device cohorts, test larger foundation models under the same calibration- and decision-aware protocol, and develop online per-subject recalibration, principled selection of the calibration set, and utility-elicited decision thresholds.

Conclusions

Calibration, not architecture, limited decision-useful cross-subject wearable stress detection across three Empatica E4 datasets and for the compact architectures evaluated here: discrimination was dataset-dependent, a compact 1D-CNN with or without self-supervised pretraining did not beat a calibrated baseline, calibration degraded under subject shift and global recalibration did not transfer, and uncalibrated probabilities carried limited or negative decision value. Few-shot per-subject recalibration reduced miscalibration and improved net benefit when calibration windows were sampled across the recording, but not under a prospective, chronological protocol, where within-recording drift prevented transfer; its benefit is therefore an upper bound rather than a prospective guarantee. Deployable wearable stress detection should therefore be evaluated in terms of calibration and decision-curve net benefit, datasets should retain subject identifiers and adequate per-subject data, and a small subject-specific calibration set is a ractical requirement. These conclusions, from small single-family cohorts with within-dataset analyses, should be confirmed in larger multi-device prospective studies. This study introduces decision-curve analysis to wearable stress detection and relocates the bottleneck from architecture to calibration under subject and base-rate shift for the compact models studied.

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Author contributions A.A. is the sole author. A.A. conceived and designed the study, implemented the analysis pipeline and software, ran the experiments, analysed and interpreted the results, prepared all figures and tables, and wrote and revised the manuscript. A.A. read and approved the final manuscript.

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Data availability The three datasets analysed in this study are publicly available from their original sources: WESAD [1], the Nurse Stress dataset [2], and the Exercise-Stress dataset [3]. The analysis code and the scripts that regenerate every figure and table in this article are openly available at https://github.com/KMDR82/wearable-stress-calibration.

Declarations

Ethics approval and consent to participate This study analysed publicly available, de-identified datasets; ethical approval and participant consent are documented in the original dataset publications [15–17]. No new human data were collected, and no additional ethical approvals were required.

Consent for publication Not applicable.

Competing interests The authors declare no competing interests.

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